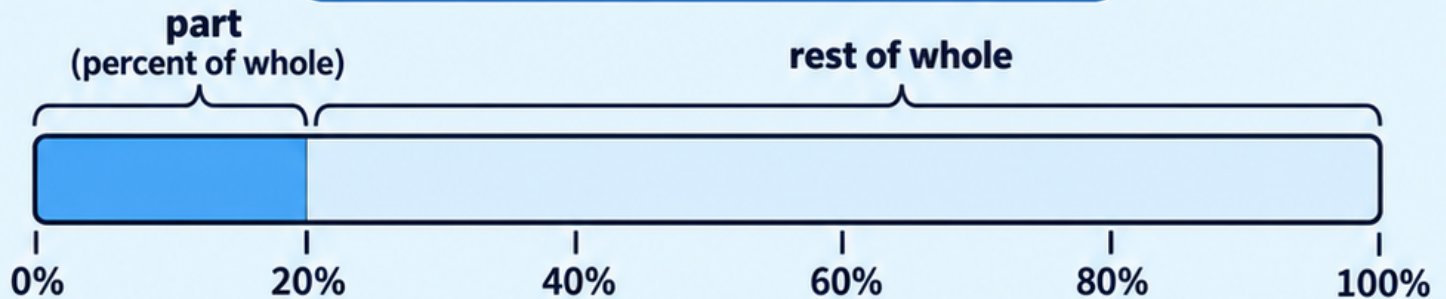


Percent Problems

Part • Whole • Percent

$$\text{part} = \text{percent} \times \text{whole}$$



The bar shows 20% of the whole.

1 Percent of a number

Find a part given a percent and a whole.

Example:

$$\begin{aligned} &20\% \text{ of } 50 \\ &= 0.20 \times 50 \\ &= 10 \end{aligned}$$

✓ 20% of 50 is 10.

2 Percent change

$$\frac{\text{new} - \text{old}}{\text{old}} \times 100$$

Example:

A price increases from 40 to 50.

$$\begin{aligned} &\frac{50 - 40}{40} \times 100 \\ &= \frac{10}{40} \times 100 \\ &= 25\% \end{aligned}$$

✓ The price increased by 25%.

3 Tax / tip / discount

$$\text{total} = \text{price} \pm (\text{percent} \times \text{price})$$

Example:

A \$20 meal with an 18% tip.

$$\begin{aligned} \text{Total} &= 20 + (0.18 \times 20) \\ &= 20 + 3.60 \\ &= \$23.60 \end{aligned}$$

✓ The total is \$23.60.

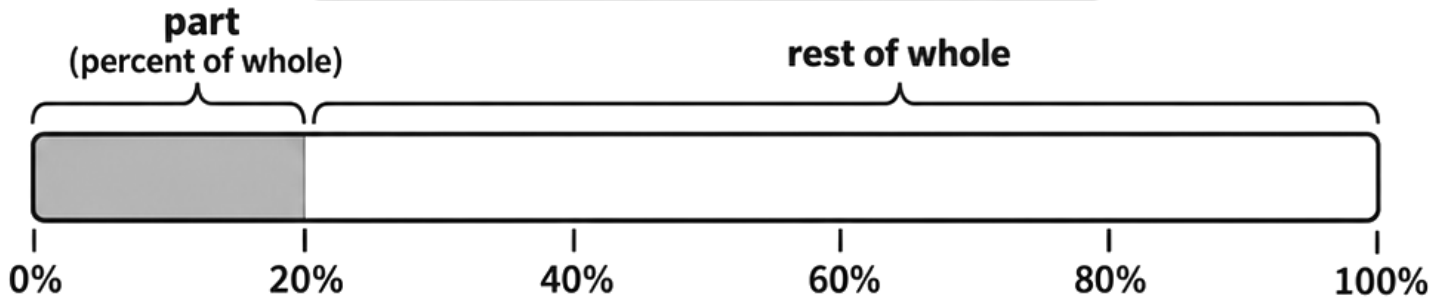


Turn the percent into a decimal before you multiply.

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